Chapter 1 · Solutions

Try each question before reading this guide. The hint gives the first move. The worked steps show why the answer follows. Letters I, N and E in the verification log stand for ID, Name and Email.

P01 · Keys and counts

Hint. Start with the declared identifiers. Add other columns to make superkeys.

The candidate keys are {ID} and {Email}. Their supersets are {ID}, {Email}, {ID,Name}, {ID,Email}, {Name,Email}, and {ID,Name,Email}. That is six superkeys. {Name} is not guaranteed to be unique in every permitted instance.

The degree is 3 and the cardinality is 4. After a fifth row, the cardinality becomes 5. The schema still has three attributes, so its degree remains 3.

Check. Each listed superkey contains at least one candidate key. A current coincidence cannot establish a schema rule.

P02 · Foreign-key changes

Hint. Ask whether any remaining child points to a missing parent.

A and B are safe for the named foreign key. Adding parent 3 does not remove any referenced value. Deleting child 10 removes a reference.

C fails because parent 9 does not exist. D fails because child 11 still references parent 2. Under ON DELETE SET NULL, D succeeds: child 11's reference changes to NULL. The final child rows are (10,1), (11,NULL), (12,NULL).

Check. SET NULL also requires a nullable child column. These conclusions do not guarantee compliance with unrelated constraints.

P03 · Join counts

Hint. Count the matches of each purchase separately.

The references 1,1,2,NULL have match counts 1,1,1,0. The inner join has 3 rows. The left join preserves the unmatched purchase and therefore has 4 rows.

After adding reference 3, the match counts are 1,1,1,0,1. The inner join has 4 rows and the left join has 5. Customer 1 still occurs twice in the inner join. A join returns matching pairs, not one row per customer.

Check. Uniqueness on the parent prevents a purchase from matching several parents. It does not prevent several purchases from matching the same parent.